

(GIT)-SOURCE CODE MANAGEMENT TOOL

❖ Introduction to SCM (Source Code Management) Tool

- **SCM (Source Code Management)** refers to a set of tools and practices used to **track, manage, and control** changes to **source code** during software development.
- SCM tools allow developers to **collaborate**, maintain code history, and ensure version control.

➤ Why use SCM?

- Tracks every change made to the code.
- Enables collaboration among multiple developers.
- Helps revert to previous versions when needed.
- Ensures code integrity.

▪ Popular SCM Tools:

- *Git*
- *Bitbucket*
- *SVN (Apache Subversion)*
- *Mercurial*
- *Perforce*

▪ Difference Between Centralized VCS and Distributed VCS

Feature	Centralized VCS (CVCS)	Distributed VCS (DVCS)
Repository Location	<i>Single central server</i>	<i>Each user has a full local copy</i>
Examples	<i>SVN, Perforce</i>	<i>Git, Mercurial</i>
Working Without Internet	✗ <i>Not possible (no access to history)</i>	✓ <i>Fully functional (local copy)</i>

		<i>available)</i>
Speed of Operations	<i>Slower (depends on network/server)</i>	<i>Faster (local operations)</i>
Backup Dependency	<i>Single point of failure</i>	<i>Multiple backups on each user's machine</i>
Merge & Branch Support	<i>Limited or complex</i>	<i>Easy and efficient</i>
History Availability	<i>Only on server</i>	<i>Full history on every machine</i>

❖ Git Introduction

- Git is a Distributed Version Control System developed by Linus Torvalds in 2005 to manage the Linux Kernel.

❖ Why use Git?

- Fast and scalable
- Branching and merging support
- Distributed architecture
- Secure with cryptographic SHA—1 hashes

❖ Lifecycle of Git

▪ Git project workflow:

- 1. Working Directory** — Your local files.
- 2. Staging Area (Index)** — Files prepared for commit.
- 3. Local Repository** (.git folder) — Where commits are stored locally.
- 4. Remote Repository** — GitHub, GitLab,

Bitbucket, etc.

❖ Lifecycle Flow:

Sql

Edit Files + git add + git commit + git push

STEPS:-

- **Initialize Repository**
 - ✓ *git init*
- **Check Status**
 - ✓ *git status*
- **Add Files to Staging**
 - *git add <file_Name>* → Add single file
 - *git add .* → Add all files
- **Commit Files**
 - ✓ *git commit -m "Initial commit"*
- **View Commit History**
 - ✓ *git log (git log -online)*
- **Undo a Commit**
 - ✓ *git revert <commit-id>*

[GENERAL INTERVIEW QUESTIONS]

Q1. Introduction to SCM Tool or Q1. What is an SCM tool?

Q2. Difference Between Centralized VCS and Distributed VCS

Q3. What is the difference between staging and committing in Git?

- Staging (*git add*) prepares changes for commit.
- Committing (*git commit*) **saves the snapshot** to the local repository.

Q5. What does git log show?

- It shows the history of commits.

Q6. What is the difference between git revert and git restore? A:

- **git revert:** Creates a new commit that **undoes a previous** commit.
- **git restore:** Used to **discard changes** in the working directory.

Q7. How is git status useful?

- It **shows the state of the working directory** and staging area.

Q8. What is Git?

- Git is a distributed version control system designed for **speed, efficiency, and reliability** in **managing source code** changes

Q9. What are some key features of Git?

- Branching and **merging, distributed** architecture, **lightweight, fast performance, data integrity** via SHA-1.

Q7. Explain the basic lifecycle of Git.

- Modify files in the **Working Directory**
- Use **git add** to stage changes (Staging Area)
- Use **git commit** to **save changes** to the local repository
- (Optional) Use **git push** to **upload changes** to a remote repository

Q8. How do you check if Git is installed?

- Run *git --version* in the terminal.

Q9. What does git init do?

- It initializes a **new Git repository** in the current directory.

Q10. What is the purpose of git add?

- It stages the changes to be committed later.

Q11. How is git commit used?

- It commits the staged changes to the local repo with a message.
Example: *git commit -m "Initial commit"*

Q12. What does git log show?

- It shows the history of commits.